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Sri Adichunchanagiri Shikshana Trust (R.)

SJC INSTITUTE OF TECHNOLOGY

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Chikkaballapura - 562 101, Karnataka



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AAVISHKAR - 2025

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ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Project Title: “A Hybrid Neural Network for EEG-based Depression Screening Using CNN+LSTM”

Abstract

Depression is a widespread psychological disorder affecting people of all ages. Early diagnosis is crucial, and EEG signals can help assess mental health. This project proposes a fully automated Depression Detection System using a novel Hybrid Neural Network, **DepHNN**, which combines CNN for temporal learning and LSTM for sequence learning. EEG data from 34 depressed and 30 normal drug-free individuals were used. The system achieves **98.27 accuracy** and a **MAE of 0.0237**, offering an efficient, accurate, and low-complexity solution for EEG-based depression screening.

Introduction

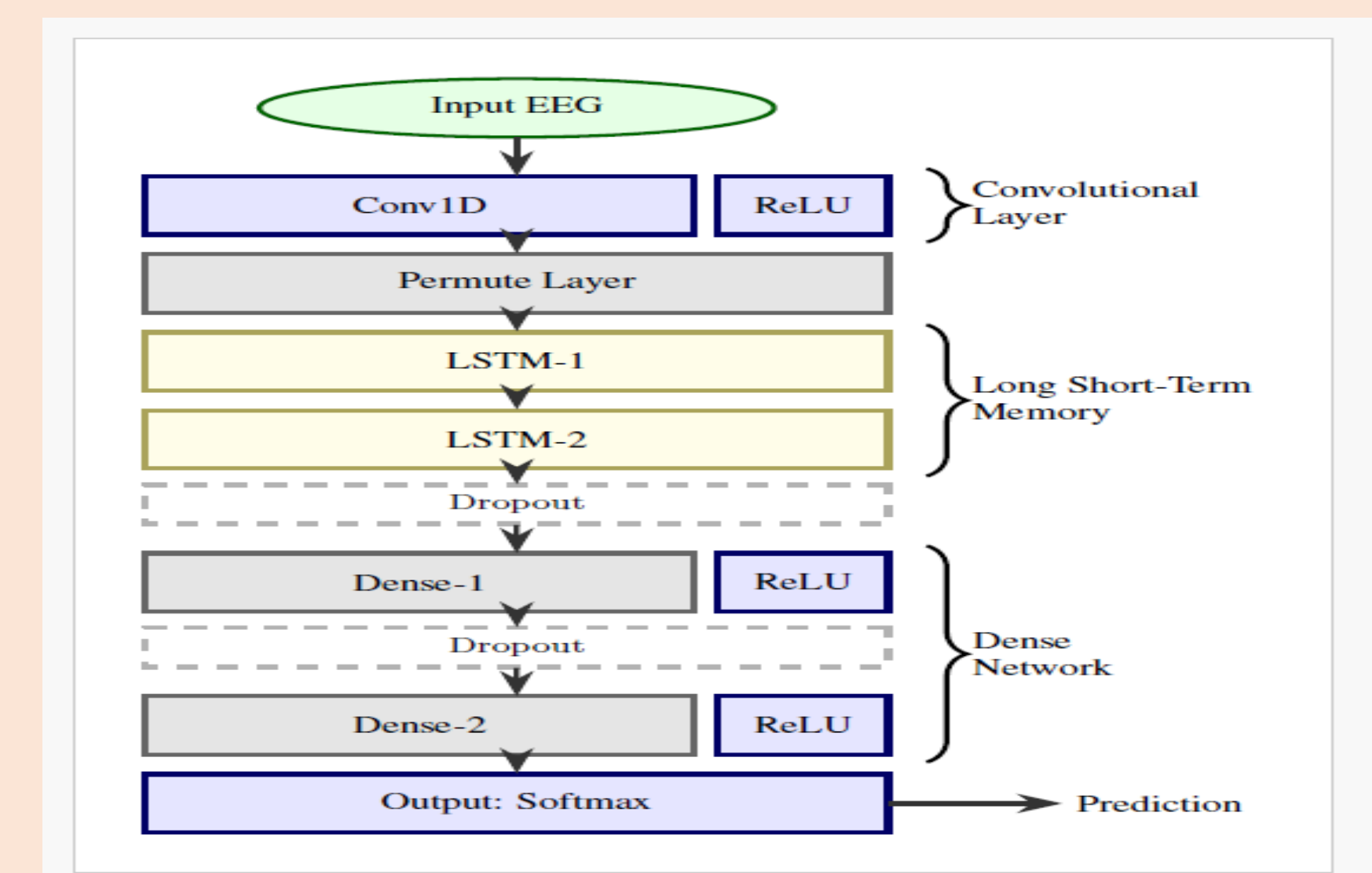
Depression affects over 322 million people worldwide, with a higher incidence among females. Although it is a treatable condition, nearly half of the cases remain undiagnosed due to factors like limited awareness, insufficient resources, and incorrect diagnoses. EEG signals, being non-invasive and cost-effective, serve as valuable biomarkers for identifying neurological disorders, including depression. However, their complexity makes manual analysis challenging. To address this, the project proposes a hybrid LSTM-CNN model (DepHNN) for efficient and accurate detection of depression using EEG data.

Objectives

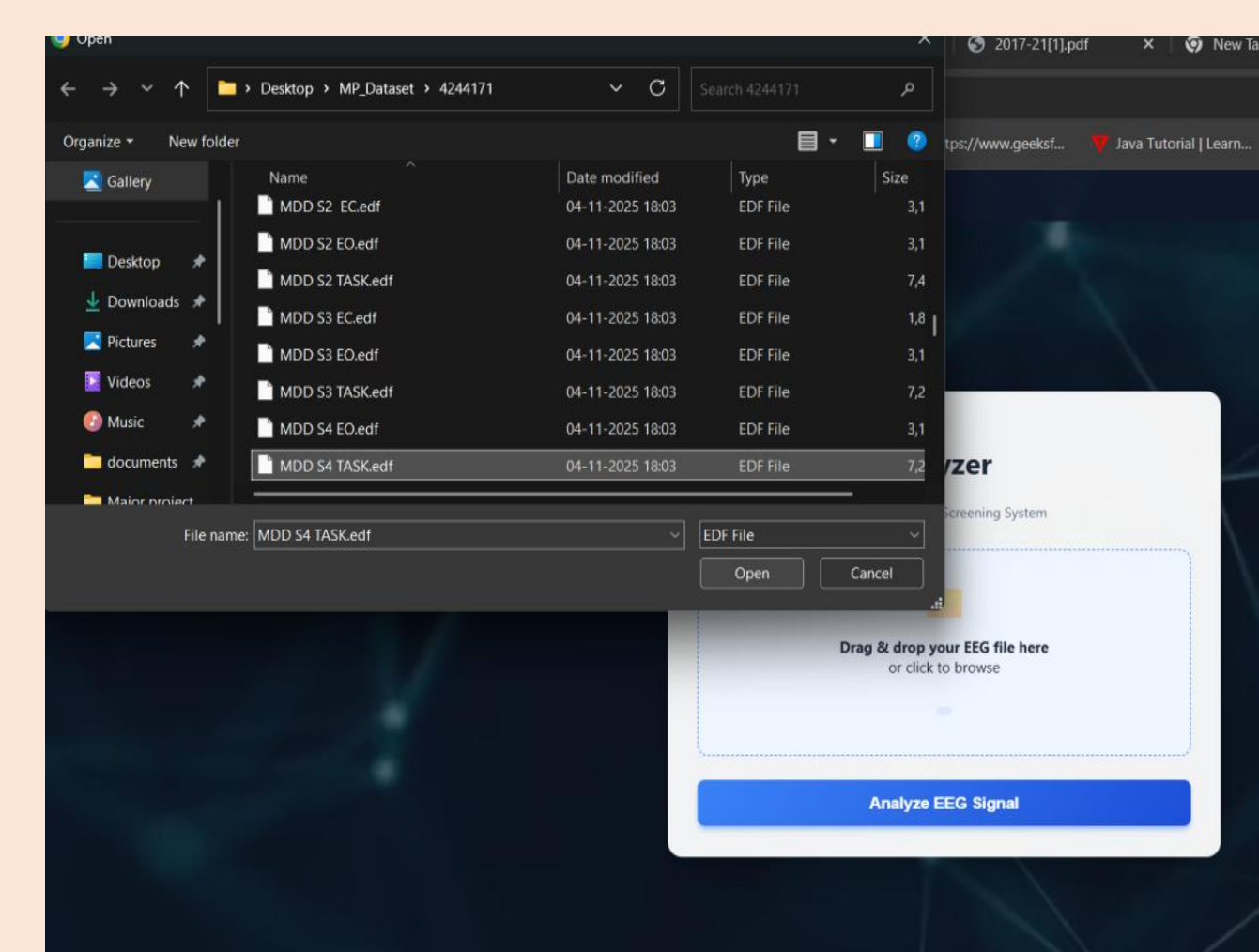
1. Improve adaptability using cross-validation and diverse datasets.
2. Apply regularization techniques such as L2 regularization and adaptive learning rate optimization.
3. Develop a CNN-LSTM model for automated depression detection using EEG signals.
4. Enhance model accuracy and generalization using dropout, data augmentation, and early stopping.
5. Minimize computational complexity and latency to enable real-time depression screening on portable and cloud-based platforms.

Methodology

- 1. Data:** EEG dataset (19-channel, eyes-closed) with **34 MDD** and **30 Healthy** subjects.
- 2. Preprocessing:** 0.5–45 Hz filtering, ICA artifact removal, Z-score normalization, 1-sec window segmentation.
- 3. Model:** Hybrid **CNN-LSTM (DepHNN)** built with TensorFlow for Healthy vs Depressed classification.
- 4. Deployment:** Django backend + secure login, responsive UI, Chart.js graphs, and real-time EEG prediction.



Results



Conclusion

DepHNN is a hybrid deep learning model combining CNN and LSTM for automated depression detection using EEG signals. It efficiently extracts temporal and sequential features without manual engineering and achieves high accuracy with minimal error and computation time. The model surpasses existing approaches in accuracy, efficiency, and robustness, making it a reliable computer-aided tool for early depression screening. Future improvements can enable severity analysis, use of larger datasets, and integration with IoMT systems for real-time mental health monitoring.

Project Batch

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